

PREDICTED CLINICAL DIAGNOSIS
SQUAMOUS CELL CARCINOMA

DIAGNOSTIC CONFIDENCE
33.4%
DenseNet-121 Softmax

1. Imaging Evidence & XAI Localization Map

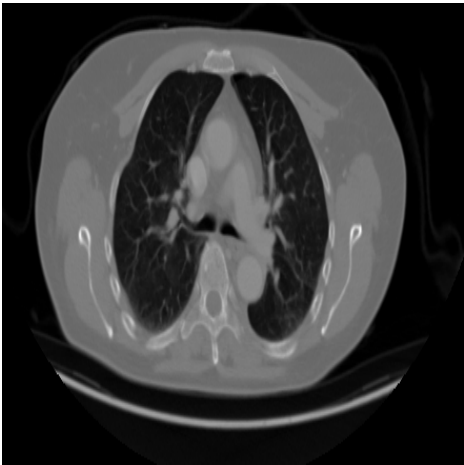


Figure 1A: Raw Input Chest CT Slice (256×256)

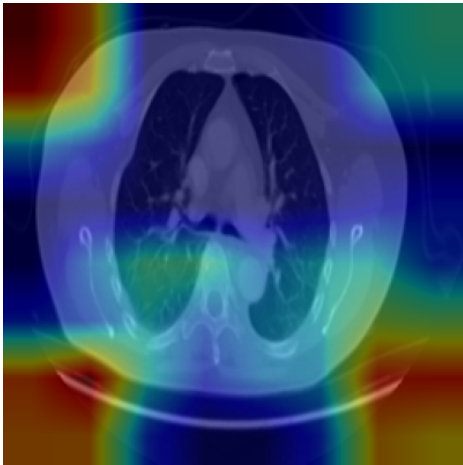


Figure 1B: Grad-CAM Activation Heatmap Overlay

2. Heatmap Feature Extraction (Quantitative Insights)

Metric / Parameter	Measurement / Value	Clinical Significance
Anatomical Location	Lower Right Lung	Primary lesion centroid in lung coordinate grid
Activation Level	High	Relative gradient magnitude on final dense layer
Region Size	7.6% of lung area	Proportion of segmented parenchyma involved
Activation Intensity	1.00 (High)	Peak normalized attribution score (0.0 – 1.0)
Spatial Distribution	Low (Diffuse)	Dispersion cluster radius around lesion epicenter

3. Natural Language Explanation (NLP Module)

Primary Finding:

"The model predicts Squamous Cell Carcinoma with 33.4% confidence. The prediction was primarily influenced by a highly activated region in the lower right portion of the lung, which occupies about 8% of the lung area."

Formal Radiological Impression:

IMPRESSION: Findings consistent with Squamous Cell Carcinoma (Left Hilum T1 N2 M0). Explainability mapping localizes the primary diagnostic determinant to the Lower Right Lung (occupying ~7.6% of segmented lung volume) with low (diffuse) concentration and peak activation intensity of 1.00. Secondary consideration: Large Cell Carcinoma (Left Hilum T2 N2 M0) (30.2%).

Recommended Next Steps:

Correlate with contrast-enhanced chest CT / PET-CT and histopathological evaluation targeting the Lower Right Lung lesion.

4. Multi-Class Probability Breakdown

Class Name	Probability	Confidence Bar
Squamous Cell Carcinoma Left Hilum T1 N2 M0 Iiia	33.41%	
Large Cell Carcinoma Left Hilum T2 N2 M0 Iiia	30.22%	
Adenocarcinoma Left Lower Lobe T2 N0 M0 Ib	25.87%	
Normal	10.50%	